

FluxTrain: An Error-Budgeted Tensor-Train Finite-Volume Solver for the Convecting Taylor–Green Vortex, with a Shared Encoding to Fault-Tolerant Quantum Solvers

Store cell averages as a quantum-style tensor network, move them with face fluxes, and account for every bit of accuracy the compression removes, so that runtime, memory and error can be plotted against Reynolds number with confidence.

Challenge: Airbus — Quantum Solvers: Enhancing Predictive Aerodynamic Modeling Capabilities

Team: Entangled Minds

Track: Tensor Network within Finite Volume (primary) + Quantum Algorithm pathway (secondary, resource-estimated)

Paradigm: Quantum-inspired tensor networks on CPU/GPU; hybrid with a Schrödingerisation-based FTQC solver analysed in emulation

Summary. Airbus asks how runtime, memory and L2 error scale with Reynolds number for the 2D convecting Taylor–Green vortex (TGV). For the robust finite-volume (FV) schemes that real high-Re flows require, resolution must grow with Re so that the dissipation the scheme adds stays below the shrinking physical dissipation. For a dense solver, memory and work then grow polynomially. We propose **FluxTrain**, a solver on the challenge's Tensor-Network track that is *strictly finite-volume*. **(A)** Cell averages of u , v and p are stored as **quantics tensor trains (QTT)**, i.e. matrix product states over the binary digits of the cell index. Interface values come from fourth-order reconstruction written as matrix product operators (MPOs), the state advances by conservative face-flux differences, and incompressibility is enforced by a discrete projection solved inside the tensor format. **(B)** A **truncation ledger** records exactly how much kinetic energy each compression step removes. We set the truncation tolerance so this “truncation viscosity” stays below 1% of the physical dissipation at every Re. That turns Re-scaling into a precision budget, and bond dimension into a measured result rather than a tuned knob. **(C)** The same QTT is, amplitude for amplitude, the state of a $2n$ -qubit register. We use the solver's measured bond dimensions and its FV operators to cost a **Schrödingerisation-based fault-tolerant solver**, emulated exactly at small size and resource-estimated up to $2^{14} \times 2^{14}$ cells. Targets: agreement with a dense FV solver to within the ledger bound; the Airbus plot of time-to-solution, memory and L2 error from $Re = 10$ to 10^8 ; $\geq 100\times$ memory reduction at 2^{10} cells per direction; and a measured runtime crossover against dense GPU FV, with controls reported whether or not they favour us.

1. Problem Framing

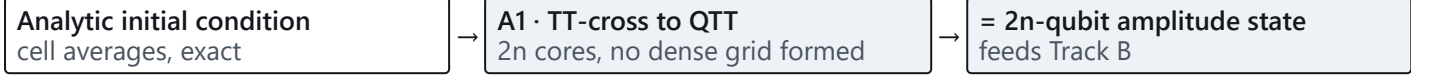
Our reading of the problem. Airbus relies on wind tunnels at the edge of the flight envelope because resolving high-Re flow is too expensive in memory and time. The TGV is a controlled proxy for that bottleneck: it has advection, pressure coupling and viscous decay, plus an exact solution against which to measure error. The request is not a single fast run. It is a *scaling law*: how cost and accuracy behave as Re, and with it the required resolution, grows. Three properties of the benchmark shape our proposal.

- **What forces resolution to grow with Re is numerical dissipation, and we measured it.** The vortex energy decays as $e^{-4\nu t}$, with $\nu = 2\pi/Re$. In our preliminary runs with an energy-conserving central scheme, the relative error in this decay rate comes from the viscous stencil alone. It is 1.3% and 1.1% at $N = 16$, and 5.0×10^{-5} at $N = 256$, *essentially the same at $Re = 100$ and $Re = 10^4$* (Appendix A3). The TGV on its own therefore does not force N to grow with Re. What does is the dissipation that robust FV schemes for real high-Re flows add through upwinding and limiters. If that numerical viscosity scales as αh^p , it stays below a fraction η of ν only if $N \propto (Re/\eta)^{1/p}$, so dense memory grows as $Re^{2/p}$ and work as $Re^{3/p}$. A tensor-network solver adds a third source, truncation, which we budget explicitly (A6). We therefore run a central and a dissipative (Rusanov) variant, and report $\nu_{\text{eff}}/\nu - 1$ alongside the L2 norm Airbus requests.
- **The “linear trap”.** For the TGV, the nonlinear term is exactly a pressure gradient: $(u' \cdot \nabla)u' = -\nabla p'$. We confirmed this numerically to machine precision (Appendix A3). The exact velocity therefore also solves the *linear* Oseen equation $\partial_t u + U_0 \partial_x u = \nu \nabla^2 u$. A solver that mishandles or even omits nonlinearity can still match the velocity field. We therefore validate the **pressure field**, which is quadratic in V_0 , and add a nonlinear stress test (§5). This matters most for quantum solvers, where linearisation is the hard part.
- **The benchmark is low-rank, and a spectral method solves it exactly.** The exact TGV cell averages have a QTT rank that does not depend on resolution: 3 or 5 depending on bit ordering, unchanged from 64^2 to 2048^2 cells, where the field needs 334–864 numbers instead of 4.2 million. A Fourier spectral solver is exact with a handful of modes. So no method can demonstrate an honest advantage over the *best* solver on TGV itself. The meaningful comparison is like-for-like: tensor-train FV against **dense FV with the identical discretisation**, the class of method Airbus uses on complex geometries. The spectral solver is shown as a reference line, and a matched-memory coarse dense grid is a mandatory control.

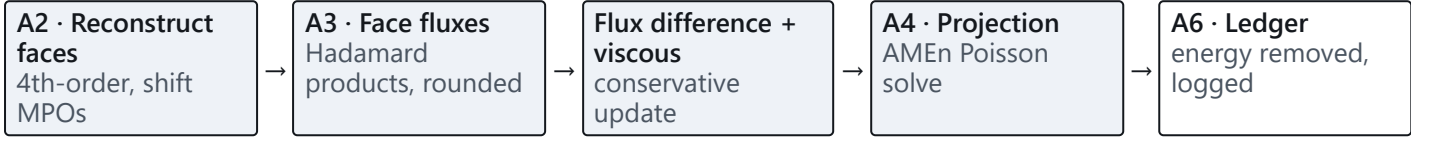
Why quantum-inspired tensor networks are a credible route. A $2^n \times 2^n$ field written over its binary digits, ordered coarse to fine, separates *scales*: each tensor core describes one level of refinement. Bond dimension χ measures how strongly those scales are correlated, which is exactly entanglement in the many-body sense. Smooth and vortex-dominated flows have bounded χ , so storage falls from N^2 numbers to about $4 \log_2 N \cdot \chi^2$. Gourianov et al. showed that χ saturates for 2D and 3D turbulent flows [20]. The recent TT-CFD literature mostly uses point-value finite differences [20–22]. The Airbus track specifically asks for *cell averages*, *interface reconstruction* and *face fluxes*, and building that FV structure into the tensor format is our first contribution. **What we do not claim:** exponential speed-up in time. Explicit time steps still scale with N . The expected gains are exponential compression of *memory*, and per-step work that grows polylogarithmically in N rather than as N^2 , provided χ stays bounded. That condition is what the PoC tests.

2. Technical Approach

SHARED ENCODING — one representation for the classical and the quantum solver



TRACK A — FluxTrain, per Runge–Kutta stage (CPU/GPU)



TRACK B — FTQC pathway (emulation and resource estimation)

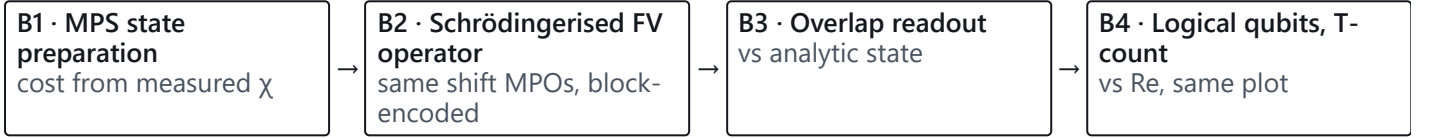


Figure 1. FluxTrain architecture. Shaded blocks operate in tensor-train format. The error against the analytic solution is computed as a tensor-train inner product, so accuracy is measured at resolutions where a dense grid could never be stored.

A. Tensor-train finite-volume solver (primary)

A1 — Cell-average encoding. Each field on $2^n \times 2^n$ cells is a QTT with $2n$ binary cores ordered by scale, either sequentially (all x digits, then y) or interleaved ($x_1 y_1 x_2 y_2 \dots$, Appendix Figure A1) [17, 20]. The choice matters. The separable TGV has rank 3 sequentially and 5 interleaved, while interleaving is expected to win once structures are genuinely 2D [20]; ablation arm 5 tests this. Unknowns are *cell averages*, not point values. The exact average of a Fourier mode is the mode multiplied by a sinc factor, so the initial condition and the analytic reference have closed forms. We build them with TT-cross [32], which samples the function rather than forming a grid, so the same step works for any initial condition.

A2 — Interface reconstruction as MPOs. The fourth-order interface value $q_{i+1/2} = (-q_{i-1} + 7q_i + 7q_{i+1} - q_{i+2})/12$ is a polynomial in the periodic shift operators $S^\pm$. Each shift is an MPO of rank 2 [18], so reconstruction is a low-rank MPO applied in one sweep. An optional Rusanov term $-1/2\alpha(q_R - q_L)$ with constant α adds controlled dissipation while keeping the operator linear. Viscous face fluxes $v(q_{i+1} - q_i)/h$ are MPOs too.

A3 — Nonlinear fluxes, conservatively. Convective fluxes such as $F^u_{i+1/2} = U_{i+1/2} \odot u_{i+1/2}$ are element-wise products of two tensor trains (rank $\chi \cdot \chi$), rounded immediately by an orthogonalising SVD sweep. The update $\bar{q} \leftarrow \bar{q} - (\Delta t/h)[(I - S^-)F + (I - S^-_y)G]$ is an exact flux difference, so the discrete sum telescopes. Rounding can perturb the domain mean by round-off, so the mean (a single TT contraction) is restored with a rank-1 correction, keeping momentum conserved.

A4 — Incompressibility. After every Runge–Kutta stage we apply a discrete projection [24, 26]: solve $L_h \phi = D_h u^*$ with the alternating minimal energy (AMEn) solver in TT format [19], warm-started from the previous stage, and subtract $G_h \phi$. Because the TGV’s pressure balances the whole nonlinear term, this step is where nonlinearity is actually tested. As an ablation, the periodic Laplacian can instead be inverted as a diagonal in QTT Fourier space [30]. The QTT Fourier transform is the classical counterpart of the quantum Fourier transform.

A5 — Time integration. SSP-RK3 [25] with $\Delta t = \text{CFL} \cdot h / |u|_{\max}$. At $\text{Re} \leq 10^2$ the viscous term is treated implicitly (Crank–Nicolson, AMEn Helmholtz solve) to remove the h^2/ν step limit.

A6 — Truncation ledger (key novelty). Compression is dissipative: rounding a tensor train removes energy. That energy is *known exactly* as $\|q\|^2 - \|q_T\|^2$ (two TT inner products) and is bounded by the sum of the squared discarded singular values [16]. Physical dissipation per step is $4\nu\|u'\|^2\Delta t$. We choose the per-step tolerance so that truncation removes at most $\eta = 1\%$ of that, giving $\tau \propto (\nu\Delta t)^{1/2} \propto (h/\text{Re})^{1/2}$. Three consequences follow. Truncation can never

mimic viscosity, even at $Re = 10^8$, where $\tau \approx 3 \times 10^{-7}$ at $N = 2^{14}$, still well within double precision. The total error splits into a discretisation part (measured against dense FV at equal N) and a truncation part (read from the ledger). And $\chi(Re)$ becomes a measured consequence of a physical accuracy requirement. The same ledger bounds how far the TT solution can drift from its dense counterpart, which is our verification criterion.

B. Quantum-algorithm pathway (secondary; emulation and resource estimation only)

B1 — Same state, costed. A normalised QTT is the amplitude vector of $2n$ qubits. Sequential circuits prepare an MPS with bond dimension $\chi = 2^b$ using a number of gates polynomial in n and χ [28]. Track A's measured χ gives the state-preparation cost directly. **B2 — Linear core by Schrödingerisation.** The FV Oseen operator $L_h = -U_c D_x + \nu \Delta_h$, built from the same shift MPOs, generates a dissipative, non-unitary evolution. Schrödingerisation [23] maps it to Hamiltonian simulation with one auxiliary register, and shifts are block-encoded as a linear combination of unitaries (quantum incrementers). Because of the linear trap, this reproduces TGV velocities exactly, which is a correctness check for the quantum circuit, *not* evidence of nonlinear capability. We say so explicitly. We emulate the algorithm exactly for $n \leq 5$ per direction (10 system and 6 auxiliary qubits), compile gates at $n = 3$ for noisy Qiskit Aer runs, and compare against Track A. **B3 — Readout without tomography.** The analytic solution has rank 5, so it can be prepared cheaply. The L2 error then follows from an overlap estimated by amplitude estimation, at $O(1/\epsilon)$ cost. **B4 — FTQC estimates.** Logical qubits and T-counts for n up to 14 are computed with Qualtran [33] and plotted on the same axes as Track A. The nonlinear cost is stated from the Carleman and quantum-LBM literature [8–10] rather than claimed.

3. Feasibility and Resource Requirements

Expected scaling (to be measured in the PoC). Here $N \propto (Re/\eta)^{1/p}$ for the dissipative variant, χ is the bond dimension and w the MPO rank:

Solver	Memory	Work to time T	Role
Dense FV, same discretisation (GPU)	$O(N^2) \propto Re^{2/p}$	$O(N^3) \propto Re^{3/p}$	Like-for-like baseline
Pseudo-spectral (Dedalus [31])	$O(N^2)$; exact for TGV at tiny N	$O(N^3 \log N)$	Reference line only
FluxTrain TT-FV (ours)	$O(\log N \cdot \chi^2)$	$O(N \log N \cdot w^3 \chi^3)$	Treatment
Schrödingerised FV, linear part	$2 \log_2 N + O(\log 1/\epsilon)$ qubits	$\tilde{O}(T \cdot (U/h + \nu/h^2) \cdot \log N / \epsilon)$	Estimate; ν/h^2 shrinks as Re grows

Preliminary evidence (already completed; Appendix A3). We built the dense FV reference that FluxTrain will be verified against: fourth-order interface reconstruction, conservative fluxes, SSP-RK3 and a discrete projection. Against the exact cell averages at $Re = 100$ and $t = 1$, the relative L2 velocity error falls from 1.8×10^{-3} ($N = 16$) to 6.3×10^{-6} ($N = 256$), an observed order of 2.0. Pressure error is 3.7×10^{-5} and the mean momentum drift is below 5×10^{-16} . The numerical solution at $N = 256$ has the same QTT rank as the exact field (5 at 10^{-8}), i.e. 564 numbers instead of 65,536, which indicates that discretisation error does not inflate rank on this problem. At $Re = 10^4$ the velocity error is lower still (6.4×10^{-8} at $N = 256$), again with rank 5. The sobering counterpart is the double shear layer at $Re = 10^4$ and $N = 256$. Its rank grows from 7 to 89 by $t = 8$ at 10^{-4} tolerance, leaving less than $2 \times$ compression at this resolution. Part of that growth is probably grid-scale noise from the non-dissipative scheme. That is why the Rusanov variant and the ledger are in the plan, and why we treat the stress test as a genuine risk rather than a formality. We also confirmed that the challenge's reference plots imply unit wavenumber on the 2π -periodic box with $\nu = 2\pi/Re$: this reproduces mean kinetic energy $0.750 \rightarrow 0.520$ at $t = 10$ and pressure amplitude 0.389 at $t = 1$, whereas reading $L = 2\pi$ inside the trigonometric terms does not.

Resource	Requirement (estimate)	Basis
Compute	≈ 350 GPU-hours on $1 \times A100/H100-80GB$; 1×32 -core CPU workstation; $\approx US\$1-1.5k$ at cloud rates; ≈ 170 kWh	Dense references to $N = 2^{14} \approx 150$ h; TT runs and ablations ≈ 150 h; stress test ≈ 50 h
Software (all open)	NumPy/CuPy, quimb (MPS/MPO, GPU back-end), ttpy (AMEn, TT-cross), Dedalus, Qiskit + Aer, Qualtran; Docker	Mature libraries; custom code is limited to the FV operators and the ledger
Data	None external: the analytic solution and challenge $\$6$ parameters	No proprietary or Airbus data needed
Quantum hardware	None required. Statevector emulation ≤ 16 qubits; noisy simulation at $n = 3$	FTQC costs by resource estimation

Assumptions and constraints. (i) The benchmark convention above is correct. We will confirm it with Airbus at kick-off, and switching convention changes only ν . (ii) Bond dimension stays bounded for the TGV under ledger-controlled rounding. The preliminary rank result supports this, and Gate G2 tests it. (iii) The scope is 2D, as specified. 3D costs are extrapolated, not measured.

Risk	Likelihood	Mitigation / fallback
Rank growth from flux products or accumulated rounding	Medium	Ledger-tuned τ ; zip-up MPO application; TT-cross evaluation of fluxes; $\chi(t)$ reported per run
TT constants too large: no runtime crossover by $N = 2^{12}$	Medium	GPU contraction; fused reconstruction and flux MPOs; the memory result stands alone; crossover reported as measured
AMEn Poisson solve stalls at large n	Low-medium	QTT Fourier-diagonal solve [30]; multigrid-style warm starts across n [29]
Multi-scale flow (shear layer) not compressible at useful accuracy	High (preliminary evidence)	Rusanov variant to suppress grid-scale noise; compression-vs-tolerance curves; claims scoped to flows where χ is measured to stay bounded
TGV too easy for TN (low rank, linear trap)	High (by design)	Pre-registered pressure metric, shear-layer stress test and matched-memory dense control
Quantum-track over-reach	Medium	Scope limited to the linear core plus resource estimates; no hardware or nonlinear claims

4. Expected Impact

What a successful PoC demonstrates. First, a tensor-network solver that satisfies the Airbus track definition *literally*: cell averages, interface reconstruction and flux-driven updates, verified against a dense FV solver to within a computed bound. Second, the plot Airbus asks for, with time-to-solution, memory and L2 error against Re from 10 to 10^8 , where dense FV stops at the largest grid that fits in memory. Third, a measured answer to *where* compressed FV overtakes dense FV, and at what error. Fourth, a quantum pathway grounded in numbers: the qubit and T-counts a fault-tolerant solver would need, computed from the same operators and measured ranks. For Airbus the relevance goes beyond TGV. The memory wall in high- Re 3D FV simulation, which drives the reliance on wind tunnels, scales as N^3 for dense grids but $3 \log_2 N \cdot \chi^2$ for tensor trains. Because FluxTrain keeps the FV building blocks (reconstruct, flux, update, project), its operators map onto the structure of production FV codes, and its error ledger gives an auditable accuracy statement for every compressed run.

Target (pre-registered)	Success threshold	Measured on / against
Verification	$\ u_{TT} - u_{dense}\ \leq 2 \times \text{ledger bound}$; observed order within ± 0.3 of dense FV	$N = 2^6 - 2^{12}$, $Re = 10^2$, 10^4
Accuracy (challenge metric)	Relative L2 velocity error $\leq 10^{-3}$ at $t = 1$ and $ v_{eff}/v - 1 \leq 1\%$, at every Re	Exact cell averages, $Re = 10 - 10^8$
Nonlinearity check	Pressure L2 error $\leq 2 \times$ the dense FV error at equal N	Exact pressure field
Memory	$\geq 100 \times$ below dense at $N = 2^{10}$; fitted memory exponent in $Re \leq 0.1$ (dense: $2/p$)	Stored parameters and peak memory
Time-to-solution	Per-step crossover at $N^* \leq 2^{12}$; $\geq 10 \times$ faster per step at $N = 2^{14}$	Dense GPU FV, same GPU, float64
Physical consistency	Discrete divergence $\leq 10^{-10}$ (relative); mean-momentum drift $\leq 10^{-12}$	Every stored step
Stress test (not low-rank)	KE and enstrophy within 1% of a converged spectral reference at $t = 8$; $\geq 10 \times$ memory reduction at that accuracy, or the failure characterised	Double shear layer [27], $Re = 10^4$
Quantum pathway	Emulated Schrödingerised solver matches the TT linear solve to $\leq 10^{-6}$ ($n \leq 5$); logical qubits and T-count vs Re	Track A operators; Qualtran estimates

Value under a negative result. If χ grows or there is no crossover, the PoC still delivers a verified open-source TT-FV solver, the measured law $N(Re)$, measurements of truncation viscosity, quantum resource estimates tied to real operators, and a note on the linear trap that Airbus can use to judge any TGV result.

5. Validation Plan

5.1 Protocol

- **Reynolds sweep.** $Re \in \{10, 10^2, \dots, 10^8\}$ using the challenge §6 parameters. Snapshots at $t = 1$ (matching the Airbus reference plots) and $t = 10$. For each method, scheme variant (central and Rusanov) and Re , N is the smallest power of two that meets *both* accuracy criteria, so every runtime is reported at matched accuracy. Dense FV runs up to $N = 2^{14}$; TT-FV runs up to $N = 2^{20}$.
- **Error metrics, fixed in advance.** Relative L2 velocity error against exact cell averages, normalised by the vortex (not mean-flow) energy, which is the stricter choice. Mean kinetic energy decay, and v_{eff} fitted from vortex energy.

Pressure L2 error. For TT runs every metric is a TT inner product with the analytic QTT, so no dense field is ever formed.

- **Cost metrics.** Wall-clock time-to-solution and time per step (median of 5 repeats, same GPU); peak memory and stored parameters; bond dimension $\chi(t)$; ledger energy. For Track B: qubits, T-count and circuit depth.

5.2 Ablation ladder (isolating the tensor-network contribution)

#	Arm	What it controls for
1	Dense FV, identical discretisation	Discretisation error: the verification reference
2	Dense FV on a coarse grid with the <i>same memory</i> as arm 6	"Just use fewer cells": key control
3	Pseudo-spectral solver	State-of-the-art reference line (exact for TGV)
4	TT-FV, fixed $\chi_{\max}$, no ledger	Value of error-budgeted truncation
5	TT-FV, sequential (x then y) bit order	Value of scale interleaving
6	FluxTrain, full A1–A6 (treatment)	—
7	Arm 6 with a QTT-Fourier pressure solve	Choice of Poisson solver
8	Linear Oseen solver, no nonlinear flux or projection	The linear trap: equal velocity error, no pressure

Compression is credited only where arm 6 beats arm 2 at matched error. On the TGV arm 2 may win, because the solution is a single mode. The double shear layer [27] is where multi-scale structure is present, and both results are reported.

5.3 Timeline and decision gates (16-week sprint)

Weeks	Work	Exit criterion
1–3	GPU port of the dense FV reference; spectral reference; analytic QTT and TT-cross metrics; benchmark harness	G1: dense FV order 2 ± 0.2 , conventions confirmed
4–7	FluxTrain A1–A6 in quimb/ttpy; verification against dense FV up to $N = 2^{12}$	G2: within $2 \times$ ledger bound; $\chi \leq 32$ at $Re = 10^4$. Otherwise switch to arm 7/TT-cross and narrow the claim to memory
8–11	Full Re sweep; crossover and memory scaling; ablations 1–8; GPU TT contraction	G3: Airbus scaling plot complete
12–14	Double shear layer stress test; Track B emulation ($n \leq 5$), noisy runs ($n = 3$), Qualtran estimates	Stress and quantum targets met or characterised
15–16	Clean-VM replication by a team member who did not write the code; open-source release; report; resource declaration	One-command reproduction of every figure

6. Hybrid / Cross-Domain Integration

Each component does what it is best at. Quantum-inspired tensor networks provide scale-separated compression and exact error accounting. Classical numerical analysis contributes conservation, stability and the projection method. GPU/HPC supplies throughput for contractions and dense references, and fault-tolerant quantum algorithms are costed where they could eventually extend the reach. **The TT solver is also the quantum solver's verifier.** Statevector emulation stops near 30 qubits, but FluxTrain evolves the same $2n$ -qubit state classically at $2n = 40$ whenever χ stays bounded. That lets us check the quantum algorithm's intended output at sizes no emulator reaches, and the measured χ tells us when a classical tensor network is enough and when a quantum register would be needed. **HPC:** the Re sweep is embarrassingly parallel and contractions run through CuPy. **We deliberately use no learned surrogates inside the solver:** the challenge rewards provable accuracy, and heuristic components would break the error ledger.

7. Team Capability

Member	Role on this project	Relevant expertise
Kyoung-Tae (Elijah) Kim MSc Quantum Computing, UPM (Spain)	Quantum Project Manager: sprint plan and gates G1–G3; benchmark harness and Re-sweep orchestration; experiment tracking and timing methodology	AI & ML, RAG, LangChain, agentic-AI orchestration and system optimisation

Member	Role on this project	Relevant expertise
Naveed Qureshi MSc Quantum Computing, UPM (Spain)	Quantum Technical Lead: QTT operators (A1–A4), truncation ledger (A6), Schrödingerisation and FTQC resource estimation (B1–B4)	Satellite mission-planning optimisation using QUBO; constrained discrete optimisation
Alejandro Valencia Blancas MSc Quantum Computing, UPM (Spain)	Quantum Solutions Architect: software architecture, dense GPU FV and spectral references, verification test suite, reproducible open-source release	Software engineering, JavaScript, database systems, mobile communications, AI; ethics and epistemology of AI

Why this team. Entangled Minds formed on the MSc in Quantum Computing at the Universidad Politécnica de Madrid (UPM). Across group projects in quantum algorithms, tensor-network and variational methods, and quantum optimisation, the three of us have worked in the way this proposal requires: design hybrid quantum–classical systems, validate in simulation first, and benchmark honestly against strong classical baselines. Naveed's experience turning mission planning into constrained optimisation carries over to rank and tolerance allocation under an error budget, and to quantum resource accounting. Elijah's AI-orchestration experience covers the 8×8 sweep of Reynolds numbers and ablation arms. Alejandro's software-engineering depth serves the verification suite and one-command reproducibility.

Execution readiness. CFD is not our core background, and we have designed the plan around that. The TGV's exact solution makes correctness self-checking, and we have already built and verified the dense FV reference (second-order convergence, conservation to machine precision). We use textbook schemes and mature libraries, and gate G2 settles the claim before the expensive sweep. We will also seek advisory input from CFD faculty at UPM's School of Aeronautical and Space Engineering (ETSIAE). Compute (≈ 350 GPU-hours, $\approx \text{US\$}1\text{--}1.5\text{k}$) will come from cloud rental or academic allocations.

References. [1]–[15] as numbered in the Airbus challenge statement §7. [16] Oseledets, SIAM J. Sci. Comput. 33 (2011) · [17] Khoromskij, Constr. Approx. 34 (2011) · [18] Kazeev & Khoromskij, SIAM J. Matrix Anal. Appl. 33 (2012) · [19] Dolgov & Savostyanov, SIAM J. Sci. Comput. 36 (2014) · [20] Gourianov et al., Nat. Comput. Sci. 2 (2022) · [21] Kiffner & Jaksch, Phys. Rev. Fluids 8 (2023) · [22] Peddinti et al., Commun. Phys. 7 (2024) · [23] Jin, Liu & Yu, arXiv:2212.13969 · [24] Chorin, Math. Comp. 22 (1968) · [25] Shu & Osher, J. Comput. Phys. 77 (1988) · [26] Almgren, Bell & Szymczak, SIAM J. Sci. Comput. 17 (1996) · [27] Bell, Colella & Glaz, J. Comput. Phys. 85 (1989) · [28] Ran, Phys. Rev. A 101 (2020) · [29] Lubasch, Moinier & Jaksch, J. Comput. Phys. 372 (2018) · [30] Dolgov, Khoromskij & Savostyanov, J. Fourier Anal. Appl. 18 (2012) · [31] Burns et al., Phys. Rev. Research 2 (2020) · [32] Oseledets & Tyrtshnikov, Linear Algebra Appl. 432 (2010) · [33] Harrigan et al., Qualtran, arXiv:2409.04643. Full citations in Appendix D.

Appendix — Supplementary Material

FluxTrain: An Error-Budgeted Tensor-Train Finite-Volume Solver for the Convecting Taylor–Green Vortex, with a Shared Encoding to Fault-Tolerant Quantum Solvers

Challenge: Airbus — Quantum Solvers: Enhancing Predictive Aerodynamic Modeling Capabilities

Team: Entangled Minds

Contents: A. Diagrams and preliminary results · B. Mathematical specification · C. Validation specifications · D. References

Supplements the 6-page concept proposal; section references (§) point to the proposal. All material comes from public literature, open-source software and the team's own work. The preliminary code (NumPy, about 300 lines) and its raw outputs will be released with the Phase 2 repository. No confidential or proprietary information is included.

A. Technical Diagrams and Preliminary Results

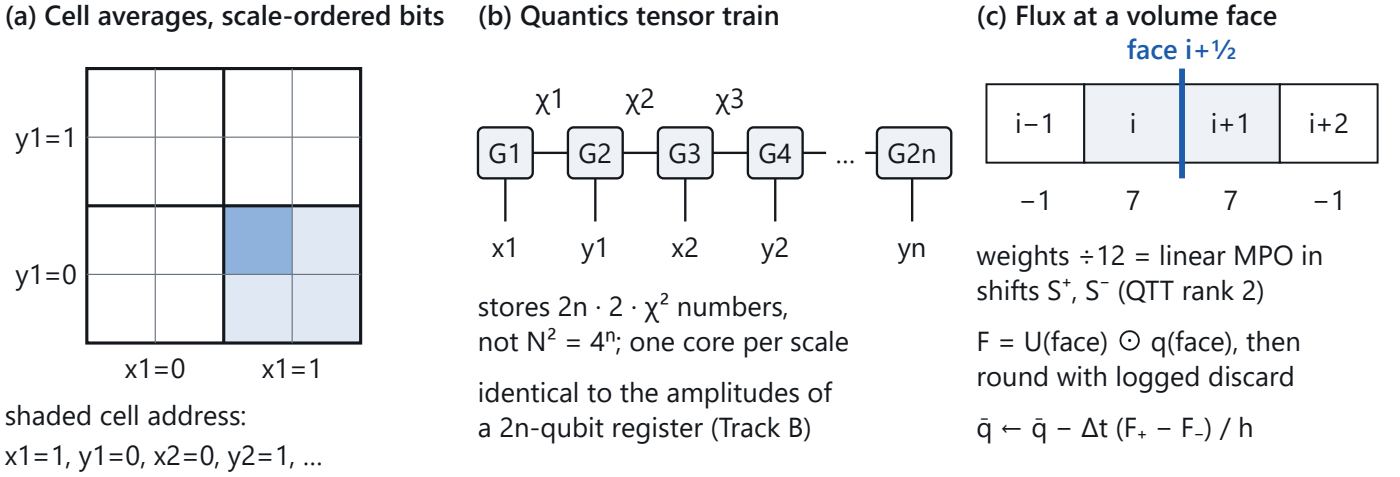


Figure A1. The FluxTrain representation. **(a)** Cell averages are addressed by binary digits from coarse to fine. **(b)** Each digit is one tensor-train core; bonds χ_k carry correlations between scales, and the array doubles as a $2n$ -qubit amplitude vector. **(c)** Interface values are fixed shift combinations (a low-rank MPO); cell averages change only through face-flux differences.

A2. Operator inventory (planning estimates)

Periodic shift $S^\pm$: binary incrementer MPO, rank 2 [18]. Fourth-order interface value $(-S^- + 7I + 7S^+ - S^+S^+)/12$: rank ≤ 4 . Flux difference $(I - S^-)/h$: rank 2; Laplacian Δ_h : rank ≤ 5 . MPO application costs $O(n \cdot w^2 \chi^3)$ with zip-up. The flux product $U_f \odot q_f$ has rank χ^2 before rounding, costing $O(n \cdot \chi^6)$ naively or $O(n \cdot \chi^4)$ with randomised rounding. The projection uses AMEn [19] to a residual $\leq 10^{-10}$. Errors, energies and the ledger are inner products with the rank-3–5 analytic QTT, $O(n \cdot \chi^2)$, with no dense field.

A3. Preliminary results (completed; dense NumPy reference on one CPU)

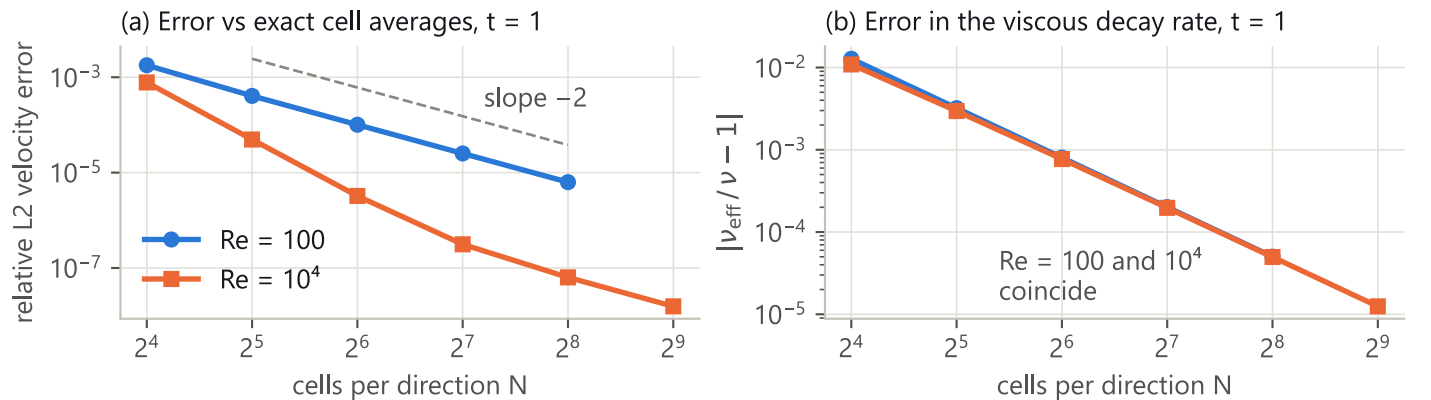


Figure A3. Dense FV reference solver (fourth-order interface reconstruction, conservative fluxes, second-order viscous flux, SSP-RK3, discrete projection) on the convecting TGV at $t = 1$. **(a)** Relative L2 velocity error against exact cell averages, normalised by the vortex energy. The observed order at $Re = 100$ is 2.14, 2.01, 2.00 and 2.00. At $Re = 10^4$ it starts near 4 (3.98, 3.93), where advective dispersion dominates, then falls through 3.37 and 2.30 to 2.02 at $N = 512$ as the second-order viscous error takes over. **(b)** The relative error in the fitted decay rate is the same at both Reynolds numbers: it equals the $h^2/12$ error of the viscous stencil, since central advection is non-dissipative. Mean momentum drift is $\leq 9 \times 10^{-16}$ in every run.

Check	Result	Why it matters
Benchmark convention: unit wavenumber on the 2π box, $v = 2\pi/\text{Re}$ ($\text{Re} = 100$)	Mean KE 0.750 / 0.694 / 0.520 at $t = 0 / 1 / 10$; pressure amplitude 0.389 at $t = 1$. Reading $L = 2\pi$ inside the trigonometric terms gives 1.085 / 0.943 / 0.093 instead	Matches Airbus plots ($0.75 \rightarrow 0.52$; ± 0.4)
Linear trap: $(u' \cdot \nabla)u' + \nabla p'$ (spectral, 64^2)	RMS 2.2×10^{-15} , against 0.50 for the advection term alone	Pressure is the nonlinearity check (§B2)
Pressure error, $N = 16 \rightarrow 512$	$\text{Re} = 100$: $1.3 \times 10^{-3} \rightarrow 3.7 \times 10^{-5}$ ($N = 256$); $\text{Re} = 10^4$: $2.8 \times 10^{-3} \rightarrow 1.3 \times 10^{-5}$	Second-order recovery
QTT rank of u (tol. $10^{-4} / 10^{-8}$)	64^2	256^2 1024^2 2048^2 Stored numbers at 2048^2 (dense 4,194,304)
TGV exact cell averages, interleaved bits	5 / 5	5 / 5 5 / 5 5 / 5 738 / 864
TGV exact cell averages, sequential bits	3 / 3	3 / 3 3 / 3 3 / 3 314 / 334
TGV dense-FV solution at $t = 1$, $N = 256$ ($\text{Re} = 100$ and 10^4)	—	— / 5 — — 564 of 65,536 at 256^2
Double shear layer [27], initial condition, interleaved	4 / 4	5 / 7 5 / 7 5 / 7 466 / 978
Double shear layer, $\text{Re} = 10^4$, $N = 256$: $t = 4$; $t = 8$	— 43 / 89; 89 / 206	— — 12,304 $\rightarrow$ 38,774 (10^{-4}) of 65,536

Reading: for the TGV, stored size grows only logarithmically with resolution, and the numerical solution is no harder to compress than the exact one. After the shear layer rolls up, rank grows sharply and compression at $N = 256$ falls below $2\times$. The central scheme is non-dissipative and the layer is marginally resolved, so some of this growth is probably grid-scale noise. Separating that noise from genuine multi-scale structure is the purpose of the Rusanov variant and the stress test (§5).

B. Mathematical Specification

B1. Benchmark as implemented, and exact cell averages

On the periodic box $[0, 2\pi]^2$ with $\rho = V_0 = U_c = 1$, $V_c = 0$ and $v = 2\pi/\text{Re}$, the exact solution is $u = U_c + V_0 \sin(x - U_c t) \cos(y - V_c t) g(t)$, $v = V_c - V_0 \cos(x - U_c t) \sin(y - V_c t) g(t)$ and $p = (\rho V_0^2/4)[\cos 2(x - U_c t) + \cos 2(y - V_c t)] g(t)^2$, with $g = e^{-2vt}$. Averaging over cell $[x_i, x_i+h]$ gives the exact expressions $S_i = [\cos(x_i - a) - \cos(x_i + h - a)]/h$ and $C_i = [\sin(x_i + h - a) - \sin(x_i - a)]/h$, so $\bar{u}_{ij} = U_c + V_0 S_i C_j g$. Each factor is a combination of two exponentials in the cell index, so the QTT rank is bounded independently of n . The mean kinetic energy is $\frac{1}{2}U_c^2 + \frac{1}{4}V_0^2 e^{-4vt}$.

B2. The linear trap

Write $u = U_{ce} + u'$. Then $(u' \cdot \nabla)u' = \frac{1}{2}(\sin 2\xi, \sin 2\zeta)g^2$ with $\xi = x - U_c t$ and $\zeta = y - V_c t$, which is exactly $-\nabla p$. The momentum equation therefore reduces to the linear Oseen equation $\partial_t u' + U_c \partial_x u' = \nu \nabla^2 u'$, which the exact velocity satisfies with no pressure at all. Velocity error alone cannot tell a Navier–Stokes solver from a linear advection–diffusion solver. Pressure error, the double shear layer, and ablation arm 8 close that gap.

B3. Update, projection and truncation ledger

Semi-discrete FV. $d\bar{u}/dt = -h^{-1}[(I - S_x^-)(U_f \odot \bar{u}_{f,x}) + (I - S_y^-)(V_f \odot \bar{u}_{f,y})] + \nu \Delta_h \bar{u} - G_h \phi$, and likewise for v . Each SSP-RK3 stage is followed by the projection $u \leftarrow u - G_h L_h^{-1} D_h u$. **Ledger.** The fluctuation $q' = q - \text{mean}(q)$ is rounded; the mean is carried exactly. Rounding to q'_τ removes energy $E_\tau = \frac{1}{2}h^2(\|q'\|^2 - \|q'_\tau\|^2) \leq \frac{1}{2}h^2 \Sigma_k \delta_k^2$ [16]. Physical vortex dissipation over one step is $4\nu E' \Delta t$. Requiring $\Sigma_{\text{stages}} E_\tau \leq \eta \cdot 4\nu E' \Delta t$ gives the relative tolerance $\tau^2 \leq 4\eta \nu \Delta t$ (per stage, after dividing by the number of roundings). Example: $\text{Re} = 10^8$, $N = 2^{14}$, $\Delta t = 0.1h$, $\eta = 0.01$ gives $\tau \approx 3.1 \times 10^{-7}$. **Dissipative variant.** A Rusanov jump on fourth-order states acts as a hyperviscosity, $\nu_{\text{num}} \approx c h^3$, so $N \propto (\text{Re}/\eta)^{1/3}$. We fit c at $\text{Re} = 10^2$ – 10^4 and test the law to 10^8 .

B4. Schrödingerisation of the FV Oseen operator (Track B)

For $du/dt = Au$ with $A = -U_c D_x + \nu \Delta_h$, write $A = -H_1 + iH_2$ with $H_1 = -(A + A^\dagger)/2 \geq 0$ and $H_2 = (A - A^\dagger)/(2i)$. With central D_x (skew-symmetric) and symmetric Δ_h , *advection is exactly the Hermitian part and viscosity exactly the dissipative part*. The warped phase $v(t, p) = e^{-|p|t}$ obeys $\partial_t v = H_1 \partial_p v + iH_2 v$ for $p > 0$. A Fourier transform in p gives $\partial_t \hat{v} = i(\eta H_1 + H_2)\hat{v}$, a Hamiltonian evolution for each η [23], and u is recovered from v at $p \geq p^\diamond$. Qubits: $2n$ (system) + n_p (auxiliary) + ancillas for the linear-combination-of-unitaries block encoding of $S^\pm$. Emulation ($n \leq 5$, $n_p = 6$) is exact, because the problem is block-diagonal in η .

C. Validation Specifications

C1. Metric definitions (fixed before any results)

- **Velocity error** $e_u = \|\bar{u}_h - \bar{u}_{ex}\|_2 / \|\bar{u}_{ex} - U_c\|_2$ (both components, over cell averages). **Pressure error** $e_p = \|p_h - p_{ex}\| / \|p_{ex}\|$, zero-mean. **Decay-rate error** $|v_{eff}/v - 1|$ with $v_{eff} = -\ln(E'(T)/E'(0))/(4T)$.
- **Consistency**: relative discrete divergence $\|D_h u\|/\|u'\|$; mean-momentum drift; cumulative ledger energy ΣE_τ as a fraction of physical dissipation.
- **Cost**: wall-clock to T (median of 5, float64, same GPU); time per step; peak memory; stored parameters; $\chi(t)$. **Crossover** N^* is the smallest N at which TT time per step is below dense time per step. **Memory exponent**: least-squares slope of $\log(\text{memory})$ against $\log(\text{Re})$ at matched accuracy.
- **Stress test**: double shear layer ($\delta = \pi/15$, $\varepsilon = 0.05$), $\text{Re} = 10^4$, $t = 4$ and 8 , against Dedalus [31] at 1024^2 (checked at 2048^2): KE and enstrophy errors, compression against tolerance.

C2. Draft Resource Declaration

Item	Declared (planning estimate; finalised in the Phase 2 report)
Hardware, GPU-hours	1× A100/H100 80GB (cloud), 1× 32-core CPU. ≈ 350 GPU-h: references ≈ 150 , TT-FV sweep and ablations ≈ 150 , stress test ≈ 50
Quantum resources	No hardware. Statevector emulation ≤ 16 qubits; Qiskit Aer noisy runs at $n = 3$ (6 system qubits); FTQC estimates with Qualtran. Exact library versions pinned in a lock file
Energy	≈ 170 kWh ($400 \text{ W} \times 350 \text{ GPU-h} \times \text{PUE } 1.2$), plus CPU ≈ 40 kWh

D. References

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- Software: NumPy, CuPy, quimb, ttpy, Dedalus, Qiskit and Qiskit Aer, Qualtran. All open-source.